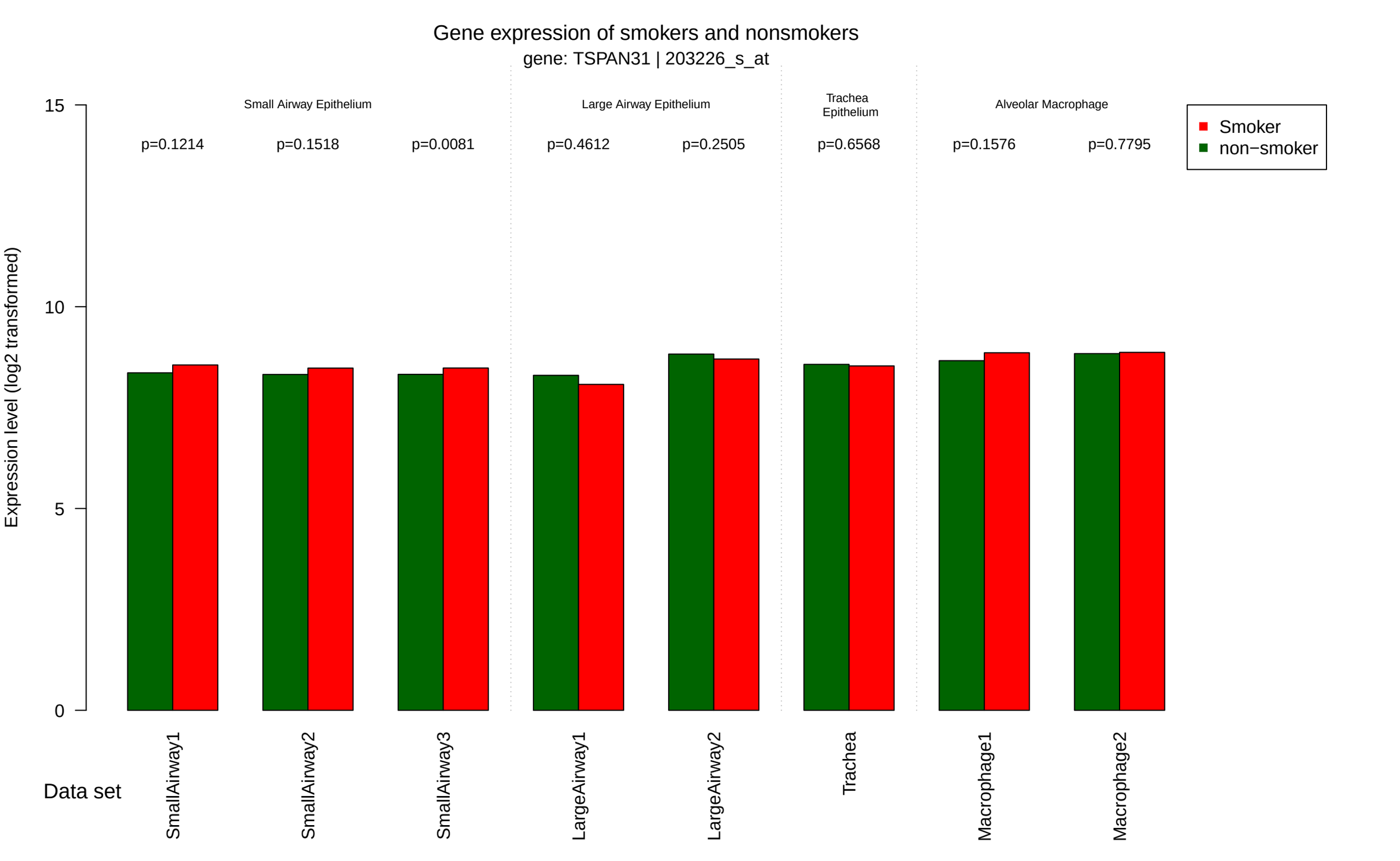
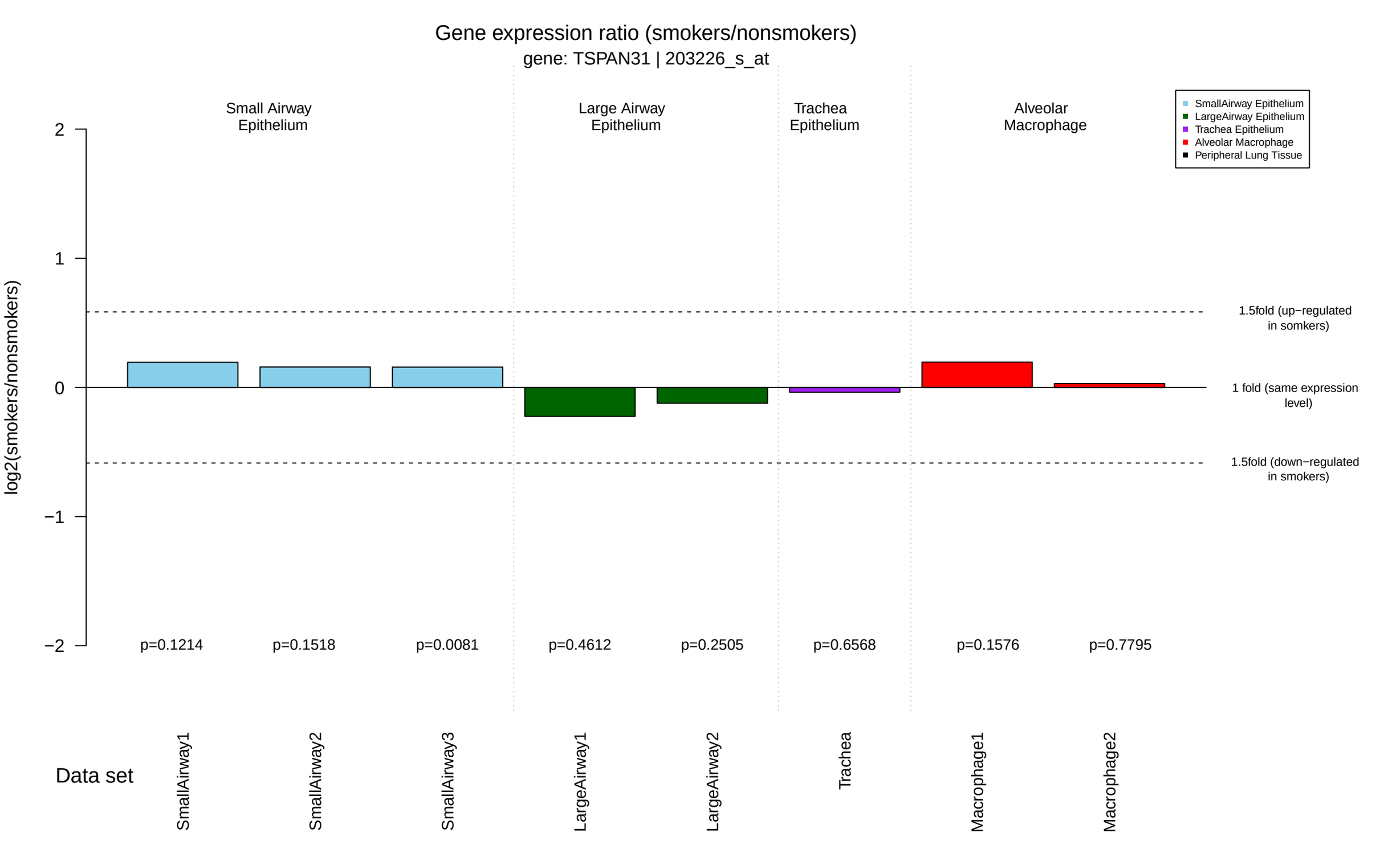
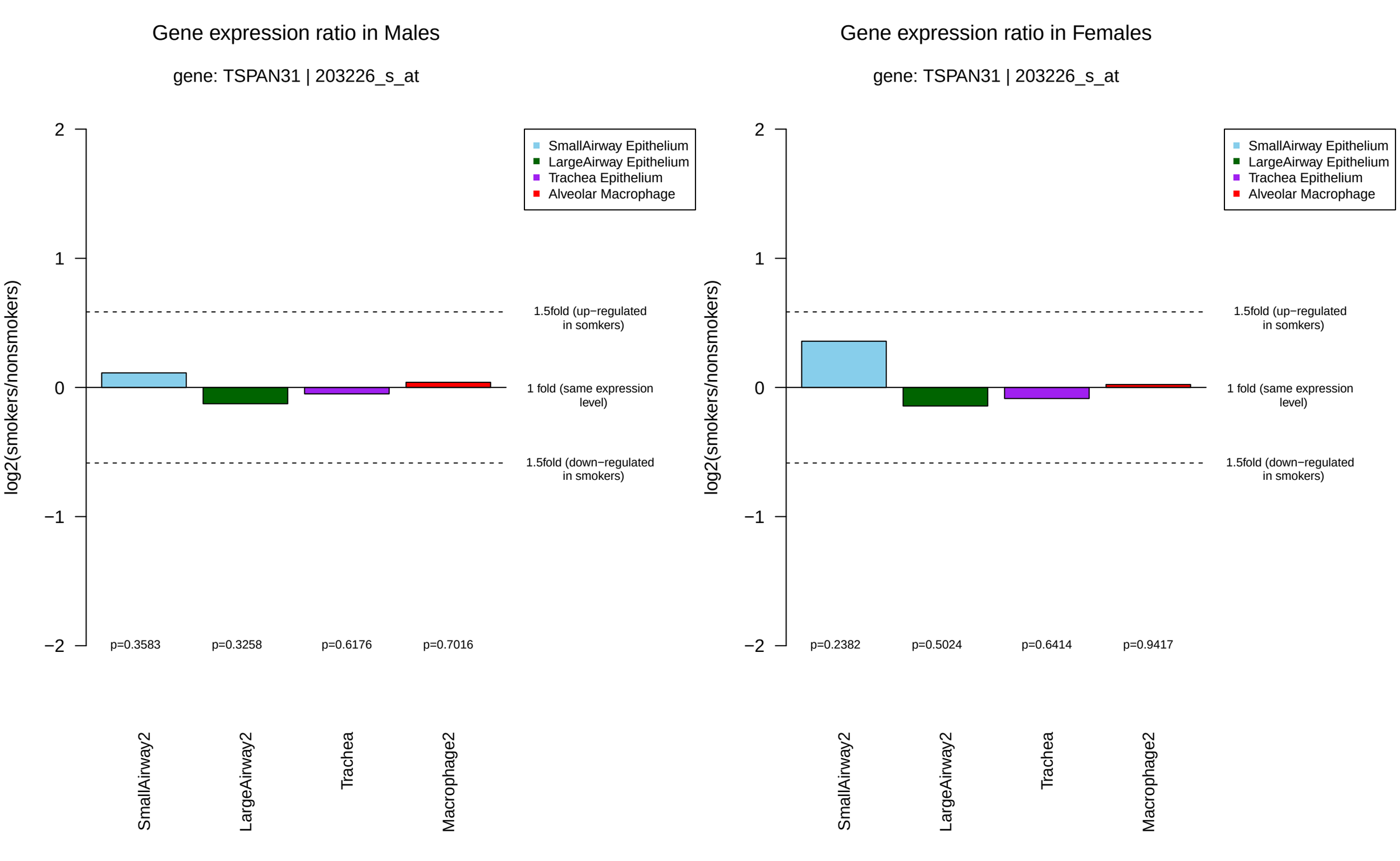
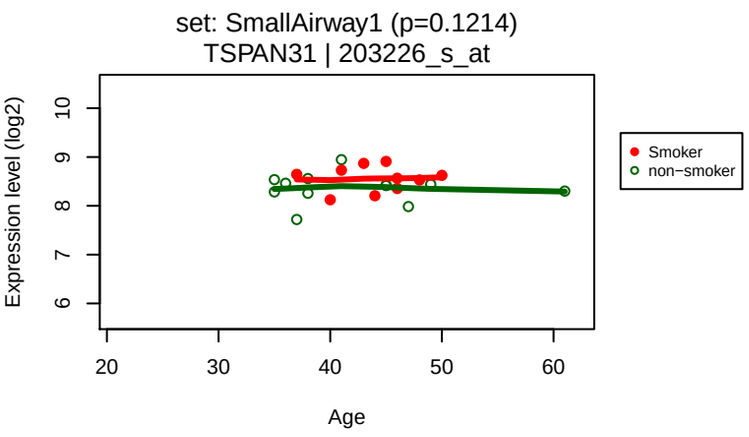
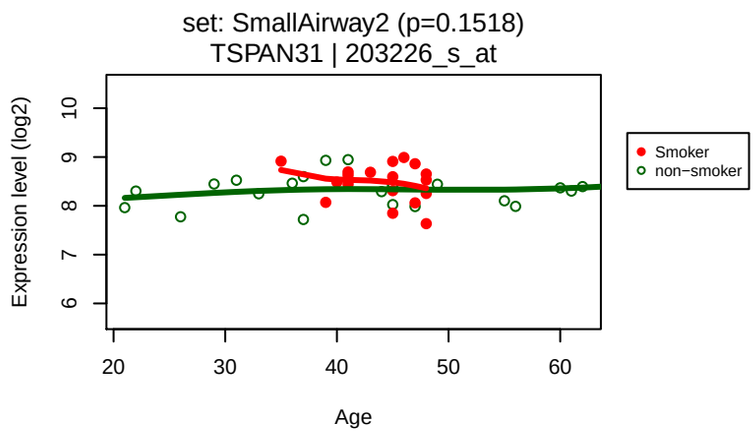
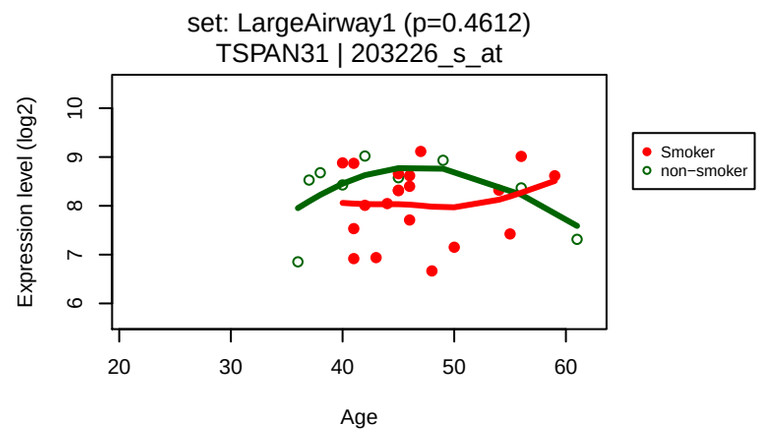

Gene expression of smokers and nonsmokers
gene: TSPAN31 | 203226_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1214
p=0.1518
p=0.0081
p=0.4612
p=0.2505
p=0.6568
p=0.1576
p=0.7795
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

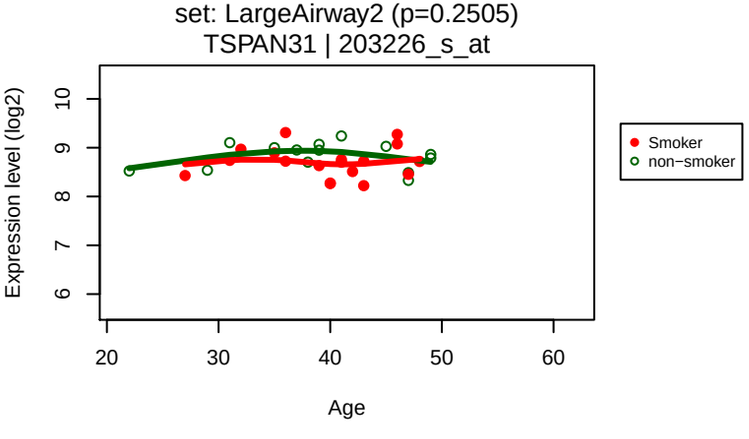

Gene expression ratio (smokers/nonsmokers)
gene: TSPAN31 | 203226_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1214
p=0.1518
p=0.0081
p=0.4612
p=0.2505
p=0.6568
p=0.1576
p=0.7795
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

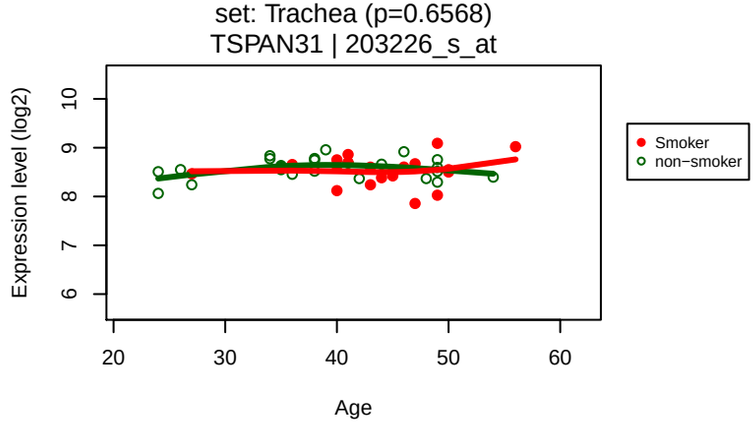

Gene expression ratio in Males
gene: TSPAN31 | 203226_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3583
p=0.3258
p=0.6176
p=0.7016
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TSPAN31 | 203226_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2382
p=0.5024
p=0.6414
p=0.9417
SmallAirway2
LargeAirway2
Trachea
Macrophage2

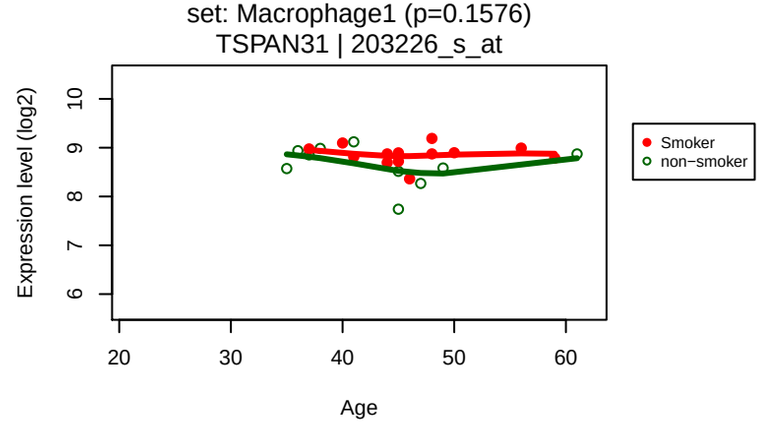

set: SmallAirway1 (p=0.1214)
TSPAN31 | 203226_s_at
Expression level (log2)
Age
Smoker
non−smoker

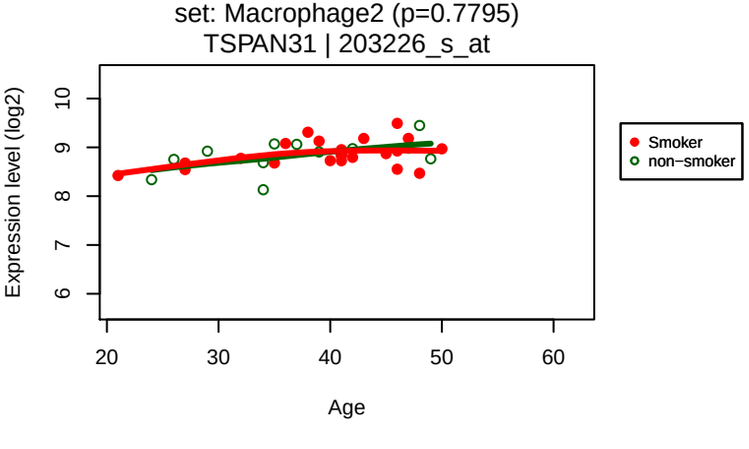

set: SmallAirway2 (p=0.1518)
TSPAN31 | 203226_s_at
Expression level (log2)
Age
Smoker
non−smoker

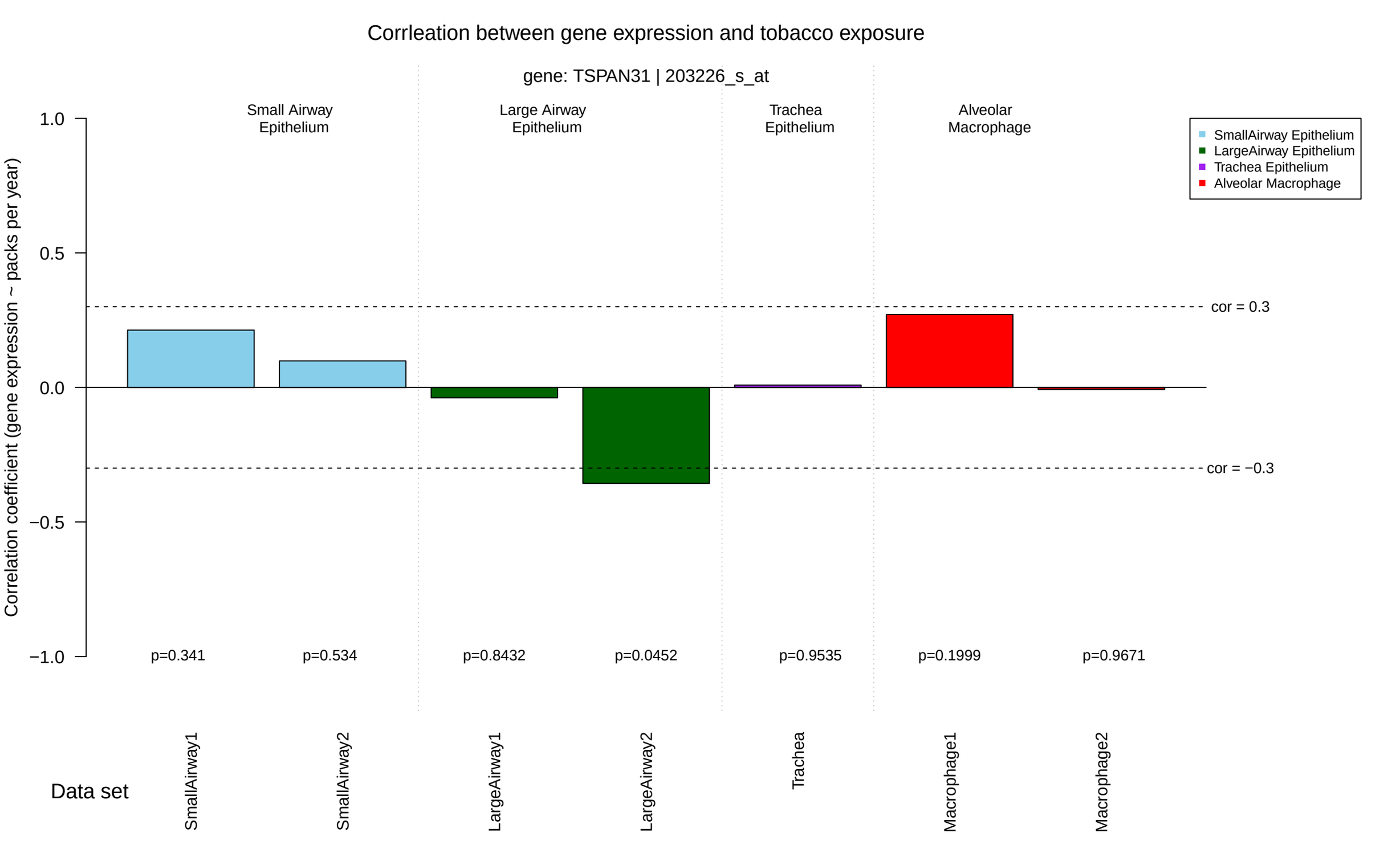

set: LargeAirway1 (p=0.4612)
TSPAN31 | 203226_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2505)
TSPAN31 | 203226_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6568)
TSPAN31 | 203226_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1576)
TSPAN31 | 203226_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7795)
TSPAN31 | 203226_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TSPAN31 | 203226_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.341
p=0.534
p=0.8432
p=0.0452
p=0.9535
p=0.1999
p=0.9671
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2